

	$p = 1.4 \times 10^5 \text{ Pa}$	A1
4ai	Frequency of the system when it is oscillation <u>freely</u> .	B1
4aii	When forced / driving frequency equals natural frequency of vibration, maximum amplitude of vibration of the oscillating body occurs.	B1 B1
4b	(What is vibrating) eg. vibration of quartz, (Why it is useful) eg. for accurate timing.	M1 A1
4ci	Resonant frequency = 35 Hz Period of oscillation, $T = 1/35 = 0.028571 \text{ s}$ Time interval = $\frac{1}{4} T = 0.00714 \approx 0.0071 \text{ s}$	C1 C1 A1
4cii	Same starting point and all points are lower. The 2 graphs should not intersect except at the starting point. Peak at lower frequency, flatter	B1 B1

